

When Does GRPO Help Budget-Controlled Summarization? Content-Gated Post-Training for Vietnamese 4B Models

Anonymous ACL submission

Abstract

Controllable summarization requires a model to preserve reference content while obeying explicit length and sentence budgets. We ask when lightweight post-training can improve this quality–control trade-off, and when group-relative policy optimization (GRPO) is useful beyond supervised instruction tuning. Our method is a reproducible recipe: instruction-conditioned LoRA-SFT, SFT-initialized group-relative updates, and a content-gated safeguard package. On the single-document test sets VietNews and WikiLingua, the final SFT + GRPO v5 systems substantially improve over their raw backbones and compare favorably to several historical baselines; the Instruct system reaches 29.62 ROUGE-2 on VietNews, and the Base system reaches 26.35 on WikiLingua. Historical VietAI, GPT, Qwen3-14B, and other baselines are retained as a clearly separated contextual benchmark until protocol-matched generations are available. We further audit the failure modes targeted by the safeguard package, including repetition loops and punctuation padding. The task uses reference-derived budgets, so the study does not claim arbitrary user-budget control; prompts, generations, and the baseline-evaluation contract are released for reproducibility.

1 Introduction

Controllable summarization is useful when systems must meet strict output budgets, such as headline generation, short mobile previews, or template-bound reporting. In these settings, a model must preserve salient content while also satisfying an explicit length target. That combination is still difficult for lightweight open-weight language models, especially in Vietnamese, where controllable summarization resources and post-training analyses remain limited.

This paper studies a practical post-training recipe as a controlled empirical case study, not as a

proposal for a new optimizer. The two instantiated systems use Qwen3-4B Base and Instruct backbones with instruction-conditioned LoRA-SFT followed by a custom group-relative update.

The central problem is to improve lexical content overlap and budget adherence together. We therefore ask four questions: (RQ1) how the final systems compare with raw controls and established external baselines; (RQ2) how much each post-training stage contributes; (RQ3) whether GRPO benefits from an SFT warm start; and (RQ4) whether the reward and decoding safeguards reduce constraint-driven degeneration.

The paper makes three contributions. First, it presents a transparent content-gated SFT-to-GRPO recipe for Vietnamese budget-controlled summarization. Second, it provides a complete stage and initialization matrix with content, length, and sentence-control metrics under a fixed evaluation contract. Third, it separates direct project evidence from historical contextual baselines and package-level reward-robustness evidence, making the comparison boundary explicit rather than conflating incompatible runs.

2 Related Work

Vietnamese summarization benchmarks.

Vietnamese summarization has moved from benchmark construction toward stronger modeling, but the literature is still comparatively small. VNDS established an early Vietnamese news summarization benchmark, and WikiLingua added a multilingual how-to benchmark that includes Vietnamese (Nguyen et al., 2019; Ladhak et al., 2020). Modeling papers then showed that pretrained Vietnamese seq2seq backbones matter: ViT5 improved Vietnamese abstractive summarization, and VieSum/BRIO reported further gains from BRIO-style training on a Vietnamese summarization corpus (Phan et al., 2022; Lam et al., 2023). In this

line of work, however, the main objective is still generic summary quality, not explicit control over length or sentence budget.

Controllable and length-aware summarization.

Controllable summarization has progressed from architectural length conditioning to prompt-based instruction following. Early work controlled output length through decoder modifications or positional encodings, and later work generalized control to keywords or natural-language prompts (Kikuchi et al., 2016; Fan et al., 2018; Takase and Okazaki, 2019; He et al., 2022; Chan et al., 2021). More recent systems tighten the length problem further by targeting exact token or sentence counts and by benchmarking instruction-controllable summarization directly (Miculicich et al., 2023; Liu et al., 2024). Prompt-based multi-control methods and instruction-based length control show that LLMs can follow structured control text, but they also show that precise adherence is still fragile, especially outside English or under multiple simultaneous constraints (Jie et al., 2024; Juseon-Do et al., 2024). This project sits in that gap: it studies reference-conditioned Vietnamese control with coupled length and sentence targets rather than free-form user preference prompting.

Post-training for summarization behavior.

Our training recipe belongs to the broader post-training line rather than to architecture changes. LoRA makes adaptation practical by freezing the backbone and learning low-rank adapters, which is attractive when the desired behavior is narrow and the model is large (Hu et al., 2022). Summarization work with learned rewards has also shown that optimizing a better reward signal can outperform ROUGE-only training, as in learning from human feedback (Stiennon et al., 2020; Böhm et al., 2019). Recent policy-optimization methods such as GRPO further reduce the memory cost of on-policy alignment and have become common in open-model post-training; in that sense, our custom rollout-reward loop is best viewed as a lightweight adaptation of this broader family rather than a new optimizer (Shao et al., 2024). The contribution here is the application of these ideas to controllable Vietnamese summarization, not a new post-training algorithm.

Reward design and reward hacking.

Once a model is optimized against a proxy reward, the reward itself becomes a design object.

The AI safety literature uses the terms reward hacking and specification gaming for cases where optimizing the proxy produces superficially successful but substantively bad behavior (Amodei et al., 2016; Skalse et al., 2022). In summarization, reward-model work has long shown that ROUGE is an imperfect proxy for human judgment, and later analyses of reward-model overoptimization make the same point more sharply: increasing proxy reward can stop correlating with true quality (Gao et al., 2023). This literature motivates the content gate, degenerate-output filter, and multiplicative reward structure used in our GRPO setup, and it also explains why we treat the changes between the pre-safeguard and safeguarded configurations as a package-level mitigation rather than a single-component ablation.

3 Datasets

3.1 Task and Prompt Setting

Each example consists of a source document x , a control instruction c , and a reference summary y^* . The control instruction specifies a target length and sentence budget derived from the reference summary statistics, so the task is best viewed as *reference-conditioned control* rather than arbitrary user-defined control. At inference time, the model receives the source and the control instruction and produces $\hat{y}$.

The instruction schema is shared across SFT, GRPO, and evaluation. All prompts are built by the project’s data augmentation pipeline. Word counts are computed as whitespace-delimited tokens, and sentence counts are computed from end punctuation. SFT always uses the approximate constraint template family, GRPO rotates across approximate, range, and upper-bound templates to diversify the control language, and evaluation fixes the approximate template family for consistent testing against the frozen artefacts.

3.2 Dataset Inventory and Validity Boundary

Table 1 summarizes the two single-document test sets and their evidence role. VietNews and WikiLingua provide the primary evaluation evidence; VietNews retains the small duplicate caveat documented in the original report.

After preprocessing, SFT uses 111,150 samples and GRPO uses 119,942 samples. The primary claims use the two single-document test sets above. Exact processing details and the external-baseline

Table 1: Single-document evaluation suite and validity boundary used in the original project evaluation.

Dataset	Task	Test	Role	Validity note
VietNews	Vietnamese news	2,000	Primary	Small duplicate caveat
WikiLingua	How-to articles	500	Primary	Original single-document test set

evaluation contract are provided in the appendix and repository documentation. Appendix A.4 records the exact preprocessing rules and Appendix A.5 shows compact prompt examples.

4 Method

Figure 1 presents the proposed lightweight post-training pipeline for controllable Vietnamese summarization. The upper row follows the end-to-end flow from task and data through preprocessing and prompt construction, Stage 1 instruction-conditioned LoRA-SFT, Stage 2 GRPO alignment, and the resulting post-trained models. The lower panel expands the GRPO training loop and the safeguard package that constrains reward optimization. At a high level, the method first teaches a Qwen3-4B backbone to summarize under explicit control instructions, then aligns that behavior with grouped rollout optimization under content-aware rewards and anti-reward-hacking defenses. The following subsections mirror this structure and make the algorithmic contributions explicit.

4.1 Task Formulation, Preprocessing, and Prompt Construction

Each training or evaluation instance is represented as a triplet (x, c, y^*) , where x is a source document, c is a control instruction, and y^* is a reference summary. The control instruction is not arbitrary user input; it is derived from y^* after preprocessing, which truncates source and summary fields to the working context budget and computes target word and sentence counts from the reference. The task is therefore controllable summarization under reference-conditioned budgets rather than unconstrained free-form prompting.

The pipeline constructs chat-style prompts rather than plain source-target pairs. For SFT, each example contains system, user, and assistant turns, and the control language is instantiated with approximate templates such as “around X words.” For GRPO and evaluation, the assistant response is removed, the reference is stored separately for re-

ward computation or scoring, and the model must generate a summary that respects the requested control budget from the system and user turns alone. Across GRPO, the control language rotates across three template families: approximate constraints, explicit ranges, and upper bounds; evaluation fixes the approximate family for consistent testing. This asymmetric prompt construction is deliberate: Stage 1 teaches the summarization task under a stable instruction schema, while Stage 2 exposes the policy to broader control wording without changing the underlying supervision target.

4.2 Stage 1: Instruction-conditioned LoRA-SFT

Stage 1 gives the backbone its base summarization and instruction-following behavior. Each processed $\{source, target\}$ pair is converted into an instruction-following chat example, so the model learns not only to compress Vietnamese content but also to map explicit control instructions to the desired output form. In practice, this stage is responsible for the largest single improvement in instruction following and summarization quality in the project, and it produces the checkpoint used to warm-start the GRPO stage.

Both Qwen3-4B families studied in the paper, Base and Instruct, are adapted with LoRA over the attention and MLP projection layers. All canonical runs use LoRA with rank $r = 32$, scaling parameter $\alpha = 32$, and dropout 0.05. The main SFT runs use a 3,072-token context length, learning rate 5×10^{-5} , warmup ratio 0.1, two epochs, and thinking mode suppression enabled.

4.3 Stage 2: GRPO Alignment

Stage 2 is implemented as a custom rollout-and-reward loop rather than through a stock trainer abstraction. In each training step, the policy formats a batch of prompts, samples multiple completions per prompt, scores them with the rule-based reward function, normalizes the rewards within each prompt group, and applies a PPO-style policy update with KL regularization. The frozen reference model is the *initial GRPO policy*: for *Fresh* GRPO it is the pretrained backbone plus a newly attached LoRA adapter at initialization, whereas for *SFT-initialized* GRPO it is the backbone plus the SFT checkpoint. The KL term therefore anchors optimization to the exact starting policy rather than to the raw backbone.

The project studies two initialization modes.

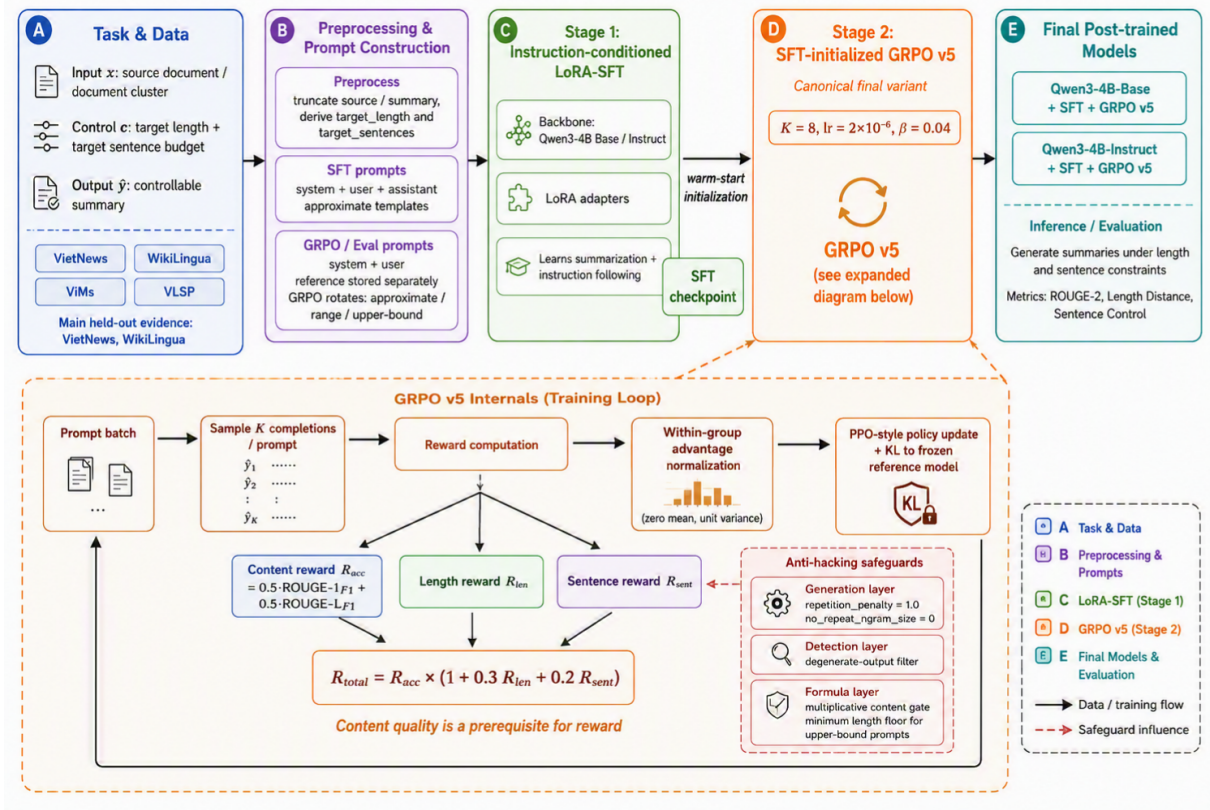


Figure 1: Overview of the proposed lightweight post-training pipeline for controllable Vietnamese summarization. The upper row summarizes task and data, preprocessing and prompt construction, Stage 1 instruction- conditioned LoRA-SFT, Stage 2 GRPO alignment, and the resulting post-trained models. The lower panel expands the GRPO training loop: prompt batches are decoded into multiple completions, rewarded for content and control adherence, normalized within each prompt group, and updated with a clipped policy objective plus KL regularization to a frozen reference model. The safeguard block summarizes the generation, detection, and formula layers used to reduce reward-hacking behavior.

Fresh GRPO attaches a new LoRA adapter directly to the pretrained backbone, while *SFT-initialized* GRPO warm-starts from the SFT checkpoint. In the final paper narrative, fresh GRPO is treated as an ablation baseline and SFT-initialized GRPO is treated as the intended deployment recipe.

Within each update, the trainer formats a batch of prompts, samples K completions per prompt, computes rewards $r_{i,k}$, and normalizes them within each prompt group:

$$A_{i,k} = \frac{r_{i,k} - \bar{r}_i}{\max(\sigma_i, 0.01)}, \quad (1)$$

where i indexes prompts and k indexes the sampled completions for prompt i . The policy is then updated with a PPO-style clipped objective over generated tokens plus a KL penalty to the frozen reference model. The per-token probability ratio is defined as

$$\rho_{i,k,t} = \exp(\log \pi_\theta - \log \pi_{\text{old}}), \quad (2)$$

and the policy objective with the clipped surrogate is

$$c_{i,k,t} = \text{clip}(\rho_{i,k,t}, 1 - \epsilon, 1 + \epsilon), \quad (3)$$

$$L_{\text{policy}} = -\frac{1}{|M|} \sum_{i,k,t} \min(\rho_{i,k,t} A_{i,k}, c_{i,k,t} A_{i,k}) \times m_{i,k,t}, \quad (4)$$

with the total loss including a KL penalty:

$$L = L_{\text{policy}} + \beta L_{\text{KL}}, \quad (5)$$

where $m_{i,k,t}$ masks generated tokens up to the first EOS token. In the implementation, π_{old} is the detached current policy within the same on-policy step, and the KL term is computed token-wise against the frozen reference model. Among the stored configurations, the final systems instantiate this stage with the canonical v5 bundle, which uses $K = 8$, learning rate 2×10^{-6} , and $\beta = 0.04$.

This makes the method a lightweight GRPO-style rollout-and-update loop tailored to the present summarization setting rather than a wrapper around a stock trainer, with Stage 2 acting as a control-refinement and failure-mitigation layer on top of the capabilities learned in Stage 1.

4.4 Reward Design and Anti-Hacking Safeguards

The reward combines content quality with explicit control adherence. Content quality is measured as

$$R_{\text{acc}} = 0.5 \text{ROUGE-1}_{F1} + 0.5 \text{ROUGE-L}_{F1}. \quad (6)$$

Length and sentence rewards, denoted R_{len} and R_{sent} , are computed from the same instruction-specific tolerance rules used to construct the control prompts in the previous subsection. The final reward uses a multiplicative gate:

$$R_{\text{total}} = R_{\text{acc}} \times (1 + 0.3R_{\text{len}} + 0.2R_{\text{sent}}), \quad (7)$$

which gates the constraint bonus on content overlap: when $R_{\text{acc}} = 0$, the total reward is zero regardless of constraint satisfaction.

Because constraint-based rewards are susceptible to reward hacking, where a model learns to satisfy length and sentence budgets with garbage content, the reward function is embedded in a defense-in-depth strategy with three layers. At the **generation** layer, the repetition penalty is kept at 1.0 and n-gram blocking is disabled (size 0). Preliminary experiments showed that higher values of either parameter, when applied by the HuggingFace decoder over the full $\approx 2,000$ -token input sequence, penalized the model for reusing source-article vocabulary and pushed decoding toward out-of-distribution symbols, which is especially damaging for short Vietnamese summaries where named entities and key terms must appear verbatim.

At the **detection** layer, a post-hoc degenerate-output filter examines each rollout with five sequential checks: empty or whitespace-only output; single-token blobs longer than 25 characters without spaces; fewer than three alphabetic characters; alpha character ratio below 12% when exceeding 15 characters; and token repetition where one token accounts for over 60% of all tokens or unique tokens fall below 30%. Any output flagged by this filter receives $R_{\text{len}} = 0$ and $R_{\text{sent}} = 0$, so degenerate rollouts cannot earn any constraint bonus; un-

Table 2: Compact nomenclature for the GRPO method variants used in the paper. Each label denotes a configuration bundle and, where applicable, a safeguard package; the labels should not be interpreted as variants differing in only one hyperparameter

Label	Defining properties	Paper role
v3 (pre-safeguard)	Linear reward; repetition penalty 1.3; n-gram blocking size 3	Reward-hacking failure baseline
v4 (historical bundle)	$K = 4$, learning rate 5×10^{-7} , $\beta = 0.15$	Bundle comparison and sentence-reward ablation
v5 (canonical bundle)	$K = 8$, learning rate 2×10^{-6} , $\beta = 0.04$; gated reward and safeguards	Main paper configuration

der the multiplicative gate, only residual content overlap captured by R_{acc} can remain.

At the **formula** layer, the multiplicative gate structurally prevents constraint-only optimization: no amount of length or sentence adherence can compensate for zero content quality. Additionally, the upper-bound constraint “không quá X từ” enforces a floor of $\max(3, 0.15 \times \max_{\text{wc}})$ words; outputs shorter than this floor receive $R_{\text{len}} = 0$, closing the loophole of satisfying the constraint with one- or two-word garbage.

4.5 GRPO Variant Nomenclature and Canonical Configuration

The paper uses a compact nomenclature for the GRPO method variants along two orthogonal axes: initialization path and configuration bundle. *Fresh* GRPO applies GRPO directly to a pretrained backbone without an SFT stage, whereas *SFT-initialized* GRPO warm-starts GRPO from the corresponding SFT checkpoint. Table 2 formalizes the compact labels used throughout the remainder of the paper.

The canonical final systems instantiate the SFT-initialized v5 branch, use 800 training steps, and keep the same LoRA adapter shape as SFT. The two final project systems differ only in their underlying Qwen3-4B backbone family: Base or Instruct. Earlier GRPO settings are retained as named historical comparison points rather than alternative deployment recipes. Appendix A.5 records the exact prompt templates and launchscript presets used to instantiate the canonical runs, while Appendix A.2 preserves the historical v4-v5

bundle comparison.

5 Experimental Results

5.1 Experimental Protocol

Primary project results use the original VietNews and WikiLingua test splits (2,000 and 500 examples), temperature 0.3, top- p 0.9, random sampling, and 256 maximum new tokens; no evaluation seed was saved. The primary tables report ROUGE-2, absolute length distance, relative length error, and sentence adherence.

The external rows below are a deliberately separated historical benchmark context. They were not regenerated under the project protocol and therefore do not support a direct superiority claim. The repository now contains an external-baseline evaluation contract; a future matched run may replace these rows without changing the RQ structure. The stored v3-v5 failure-analysis snapshot is used only for the package-level analysis and is never numerically mixed with the main project table.

5.2 RQ1: Comparative Effectiveness Against External Baselines

Tables 3 and 4 provide the headline benchmark of the paper. The two project rows are the final SFT + GRPO v5 systems, and the remaining rows are comparison baselines for Section 5.2.

The project rows substantially improve over the matched raw Qwen3 controls on both primary datasets. The external rows reproduce the historical report context requested for completeness, but they must not be read as direct wins or losses: their split, prompt, decoding, model revision, and scorer provenance are not yet matched. The repository’s baseline contract specifies the preflight checks and artifact fields required before an external row can move into the direct benchmark band.

Figure 2 makes the same comparison relative to each baseline. Positive $\Delta R2$ indicates higher ROUGE-2, while positive LenRed indicates a reduction in absolute length distance. The exact percentages are printed in the cells; color is clipped only for readability. The external-baseline cells remain contextual because those rows were not rerun under the project protocol.

The two backbone instantiations expose a useful trade-off. Instruct SFT+GRPO v5 has the higher VietNews ROUGE-2 score, whereas Base SFT+GRPO v5 is slightly stronger on WikiLingua.

We therefore retain both systems rather than collapsing the method to a single backbone.

5.3 RQ2: Stage-wise Quality-Control Trade-off

Table 5 compresses the internal project comparison into one stage-wise ablation over VietNews and WikiLingua. Its purpose is to analyze the observed effect of the two-stage post-training pipeline (SFT + GRPO v5) relative to SFT-only, pretrained-only, and fresh-GRPO variants. Appendix A.2 records the v4-v5 bundle comparison; this subsection answers the stage contribution question using the original project evaluation snapshot. To better visualize this multi-stage interaction on the primary dual objectives—retaining content quality while enforcing a strict length budget—we plot the joint ROUGE-2 and absolute summary length distance trajectory in Figure 3.

The stage-wise comparison yields three observations about the evaluated training recipe:

SFT provides the largest stage-wise gain.

Both pretrained backbones have substantially lower ROUGE-2 than their SFT variants on both primary datasets. The largest content gains occur after SFT: macro ROUGE-2 rises from 10.34 to 26.17 for Base and from 14.67 to 26.15 for Instruct. We therefore treat SFT as the dominant quality stage, with subsequent GRPO interpreted as a refinement of this strong starting point.

Warm-started GRPO improves content and, for Base, control.

SFT-initialized GRPO v5 raises macro ROUGE-2 from 26.17 to 28.45 for Base and from 26.15 to 28.81 for Instruct. For Base, it also reduces mean length distance from 9.96 to 4.76 words; for Instruct, SFT starts at 5.48 words and GRPO changes it slightly to 5.89 words. On VietNews, the Base warm-start improves ROUGE-2 from 26.61 to 28.97 and length distance from 5.73 to 1.16 words.

Fresh GRPO underperforms SFT initialization.

Applying GRPO directly to the pretrained checkpoint (Fresh GRPO v5, without the intermediate SFT step) underperforms SFT-initialized GRPO on both primary datasets. As shown by the isolated markers in Figure 3, the Fresh GRPO models reach only 20.73 and 18.67 ROUGE-2 on VietNews, compared with 28.97 and 29.62 after SFT

Table 3: ROUGE-2 (%) on the primary VietNews/WikiLingua single-document suite. Project rows use the original frozen project protocol; external rows are historical contextual values, not protocol-matched reruns, and are not ranked against project rows.

Model	VietNews	WikiLingua
<i>Project systems</i>		
Qwen3-4B-Base + SFT + GRPO v5	<u>28.97</u>	26.35
Qwen3-4B-Instruct + SFT + GRPO v5	29.62	<u>25.59</u>
<i>Raw Qwen3 baselines (frozen project protocol)</i>		
Qwen3-4B-Base (raw)	12.13	3.18
Qwen3-4B-Instruct (raw)	16.48	7.44
<i>Historical external baselines</i>		
VietAI	34.24	33.12
GPT-4o	21.61	20.65
GPT-3.5-Turbo	28.13	21.09
Qwen3-14B	18.51	20.24
Llama3.3-70B-Instruct	22.17	19.40
Phi4-14B	13.17	14.28
Sailor-20B-chat	17.70	18.74
VinBigdata (7B)	20.59	21.02

initialization. The same ordering holds on WikiLingua (9.71/8.09 versus 26.35/25.59), supporting SFT warm-starting as the reliable condition for useful GRPO in this study.

Observed trade-offs and backbone choice.

The trajectory curves show that the two-stage recipe has a genuine trade-off rather than a universal win on every control metric. Instruct SFT+GRPO v5 has the highest VietNews ROUGE-2 (29.62), while Base SFT+GRPO v5 is slightly stronger on WikiLingua (26.35 versus 25.59) and gives the lowest single-dataset length distance on VietNews (1.16 words). SFT alone gives the lowest VietNews length distance for Instruct (1.09 words), whereas warm-started GRPO gives the lowest WikiLingua distance for Base (8.35 words). The result supports a conditional-refinement claim: GRPO is most useful after SFT, but its content and control effects depend on the backbone and domain.

5.4 RQ3: GRPO Initialization and Sentence Control

The initialization comparison is a separate question from the stage decomposition above. Holding the v5 bundle fixed, SFT-initialized GRPO is consistently stronger than Fresh GRPO: pooled ROUGE-2 is 28.45 versus 18.53 for Base and 28.81 versus 16.55 for Instruct. The gap is espe-

cially large on WikiLingua (Base: 26.35 versus 9.71; Instruct: 25.59 versus 8.09), while the length-distance gap is also substantial for the Base branch. These paired differences motivate the warm-start recipe, but they do not establish that SFT initialization is the only causal factor because the two branches begin from different learned policies.

Sentence adherence is computed from the same saved generations rather than from a separate evaluation run. Exact match and mean absolute error (MAE) show that the final systems preserve control while improving content:

The Instruct branch starts with stronger sentence control; GRPO further improves VietNews exactness but slightly changes WikiLingua adherence. This is why sentence metrics are supporting evidence for the content-control trade-off rather than a single optimized objective.

5.5 RQ4: Reward Robustness and Mitigation Package

Reward hacking occurs when a model optimizes the reward signal without improving output quality. The following analysis is explicitly historical: its stored v3/v5 generations use the earlier failure-analysis snapshot, whereas the original project evaluation supplies the stage-wise results above. In v3 (the pre-safeguard bundle), three interacting design flaws created conditions under which this be-

Table 4: Absolute summary length distance (lower is better) on the primary VietNews/WikiLingua suite. Project rows use the original frozen project protocol; external rows are historical contextual values, not protocol-matched reruns, and are not ranked against project rows.

Model	VietNews	WikiLingua
<i>Project systems</i>		
Qwen3-4B-Base + SFT + GRPO v5	1.16	8.35
Qwen3-4B-Instruct + SFT + GRPO v5	<u>1.52</u>	<u>10.25</u>
<i>Raw Qwen3 baselines</i>		
Qwen3-4B-Base (raw)	33.03	61.59
Qwen3-4B-Instruct (raw)	4.01	23.42
<i>External baselines</i>		
VietAI	—	—
GPT-4o	8	11
GPT-3.5-Turbo	18	17
Qwen3-14B	50	45
Llama3.3-70B-Instruct	14	13
Phi4-14B	205	214
Sailor-20B-chat	38	43
VinBigdata (7B)	115	89

Table 5: Stage-wise comparison on the two primary held-out Vietnamese benchmarks. ROUGE-2 is reported in percent and LenDist is mean absolute length distance in words.

Model	VietNews ROUGE-2	WikiLingua ROUGE-2	VietNews LenDist	WikiLingua LenDist
Base pretrained	12.13	3.18	33.03	61.59
Base + SFT	26.61	24.43	5.73	14.18
Base + fresh GRPO v5	20.73	9.71	3.80	22.03
Base + SFT + GRPO v5	<u>28.97</u>	26.35	<u>1.16</u>	8.35
Instruct pretrained	16.48	7.44	4.01	23.42
Instruct + SFT	26.70	23.93	1.09	<u>9.87</u>
Instruct + fresh GRPO v5	18.67	8.09	2.95	16.12
Instruct + SFT + GRPO v5	29.62	<u>25.59</u>	<u>1.52</u>	<u>10.25</u>

Table 6: Sentence-control evidence on the primary held-out suite. Exact is the fraction matching the requested sentence count; MAE is in sentences.

Model	VN Exact	VN MAE	WL Exact	WL MAE
Base + SFT	91.40	4.299	58.6	1.674
Base + SFT + GRPO v5	93.85	0.316	65.6	0.470
Instruct + SFT	<u>95.45</u>	<u>0.058</u>	88.0	0.204
Instruct + SFT + GRPO v5	96.95	0.041	<u>78.8</u>	<u>0.340</u>

havior emerged. First, v3 used a **linear reward sum**

$$R_{\text{total}} = w_{\text{acc}}R_{\text{acc}} + w_{\text{len}}R_{\text{len}} + w_{\text{sent}}R_{\text{sent}},$$

which allowed the constraint terms to contribute independently of content: a model could score ≈ 0.50 by satisfying length and sentence budgets with garbage while ignoring R_{acc} . Second, a repetition

penalty of 1.3 was applied by the HuggingFace decoder over the full $\approx 2,000$ -token input sequence, penalizing the model for reusing any vocabulary from the source article and pushing generation toward out-of-distribution symbols. Third, n-gram blocking with size 3 blocked source trigrams, forbidding named entities and key Vietnamese content words. The resulting failure modes, concentrated in the Base branches where the pretrained backbone lacks the instruction-following prior of the Instruct model, included repetition loops (e.g., assistant repeated 128 times, ROUGE-2 = 0.0), punctuation padding (245 words of periods, LenErr = 1189%), digit blobs that satisfied “không quá X từ” with $R_{\text{len}} = 1.0$ and $R_{\text{acc}} = 0$, and comma-separated token loops carrying no semantic content. Figure 4 illustrates the two most severe modes qualitatively. Appendix A.2 records

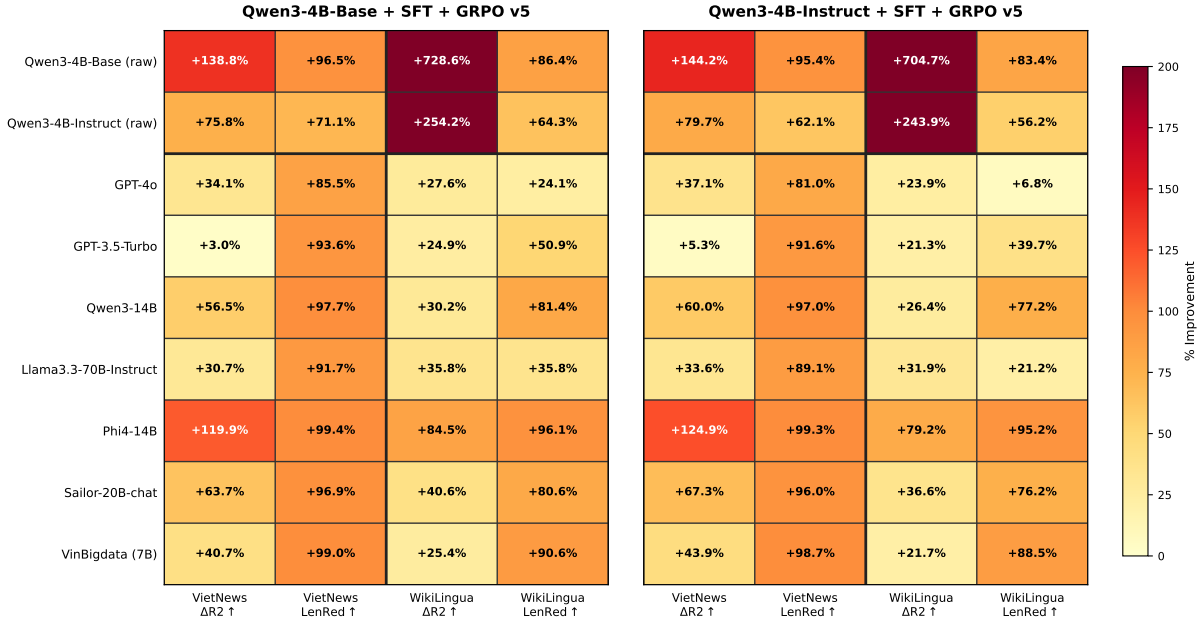


Figure 2: Baseline-relative improvement of the two final project systems on VietNews and WikiLingua. Each cell reports percentage change in ROUGE-2 ($\Delta R2$) or absolute length distance (LenRed) relative to the corresponding baseline; positive values favor the final system. Color is clipped at 200% while exact values remain annotated.

the historical bundle-level v4-v5 comparison, and Appendix A.6 keeps the sentence-reward ablation that remains inconclusive under the frozen sampling protocol.

The canonical v5 bundle addresses each root cause through the defense-in-depth safeguards described in Section 4.4: the generation-layer fixes restore natural decoding, the detection-layer filter catches degenerate outputs, and the multiplicative gate structurally prevents constraint-only optimization.

Table 7 reports the historical quantitative evidence. Base fresh shows the most dramatic recovery in that snapshot: degenerate rate collapses from 23.81% to 0.32% ($\approx 74\times$ reduction), $R_{acc} = 0$ drops from 27.16% to 1.94%, and ROUGE-2 rises from 10.58 to 18.66. Base SFT-init suffered severe length explosion under v3 (LenErr = 425%), which v5 cuts to 14.94% while raising ROUGE-2 from 18.03 to 26.38. The Instruct branches were substantially more robust under v3 (degenerate rates remained below 1%), owing to their chat-tuning prior; their v3-to-v5 gains are correspondingly incremental, with Instruct SFT-init v5 reaching the highest mean R_{acc} (0.4213) and ROUGE-2 (27.65). Within that historical snapshot, the v5 safeguard package sharply reduces catastrophic failures for the Base branches and preserves the already-low Instruct degeneration rate. This is sup-

porting evidence for the package, rather than a causal ablation of one detector or hyperparameter.

The historical comparison supports v5 as a stable canonical configuration for the main stage-wise evaluation. The Instruct branches’ near-immunity to degeneration is consistent with the primary finding that an SFT warm-start or equivalent generative prior is useful for effective GRPO in this setting. The v3-to-v5 improvement is properly understood as the effect of a **mitigation package** (generation, detection, and formula-level safeguards acting jointly), rather than an ablation of any single component. Appendix A.3 retains the sentence-control results that justify keeping sentence reward as an explicit metric, and Appendix A.6 shows that removing that term does not support a causal claim under the current evaluation protocol.

6 Conclusion

Controllable Vietnamese summarization is both a content-generation and alignment problem: the model must preserve salient information while obeying explicit length and sentence budgets. This paper studies that problem through a controlled Qwen3-4B post-training comparison. Across the primary VietNews and WikiLingua evaluation suite, SFT supplies the dominant content gain, SFT-initialized GRPO acts as a conditional refinement, and Fresh GRPO remains clearly

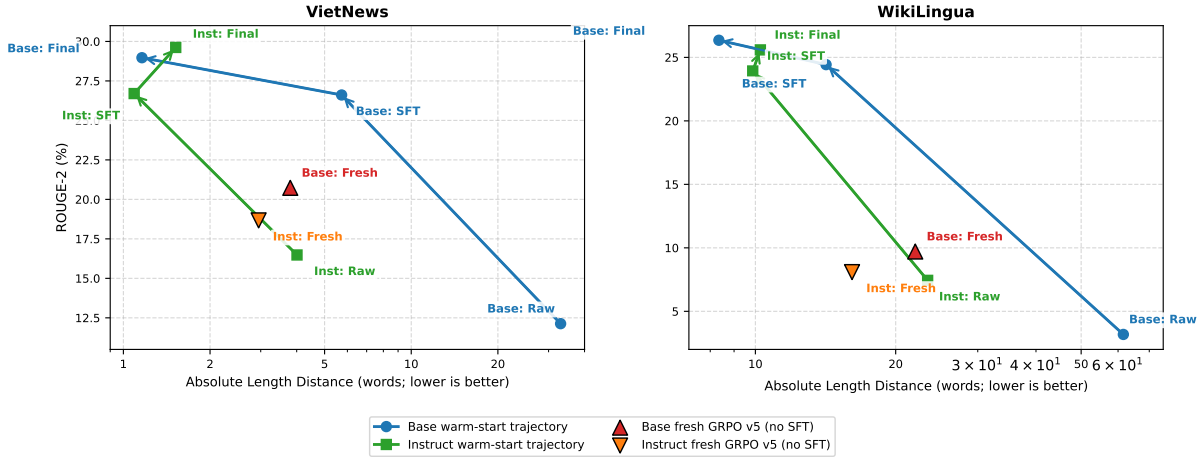


Figure 3: Stage-wise quality-control trajectory under the original project evaluation protocol. Arrowed lines show Raw $\rightarrow$ SFT $\rightarrow$ SFT+GRPO v5; standalone markers show Fresh GRPO v5 without SFT initialization.

Table 7: Historical v3-v5 reward-hacking snapshot. Deg.% is the fraction flagged as degenerate by the detection filter; $R_{acc} = 0\%$ is the fraction receiving zero content reward; Mean R_{acc} is the average content reward. This package-level comparison is separate from the main project evaluation and does not isolate any single change.

Run	ROUGE-2	LenErr%	Deg.%	$R_{acc} = 0\%$	Mean R_{acc}
Base fresh v3	10.58	163.16	23.81	27.16	0.1854
Base fresh v5	18.66	29.05	0.32	1.94	0.3160
Base SFT-init v3	18.03	425.00	14.77	1.16	0.2508
Base SFT-init v5	26.38	14.94	2.71	0.45	0.4009
Instruct fresh v3	15.00	30.08	0.03	1.48	0.2768
Instruct fresh v5	16.69	22.78	0.03	1.58	0.2979
Instruct SFT-init v3	25.06	19.94	0.94	1.19	0.3837
Instruct SFT-init v5	27.65	15.85	1.42	0.45	0.4213

weaker than warm-started GRPO. The strongest result is 29.62 ROUGE-2 on VietNews from Instruct SFT+GRPO v5; Base SFT+GRPO v5 reaches 28.97 on VietNews and mean length distances of 1.16 and 8.35 words on VietNews and WikiLingua. These conclusions are deliberately bounded to reference-conditioned Vietnamese single-document summarization under the stated original evaluation protocol. The study does not establish arbitrary-budget controllability, superiority over external models evaluated under different protocols, or a causal effect for any individual v3-v5 hyperparameter. The results also show that control gains are backbone- and domain-dependent: Instruct SFT already has strong length adherence, while Base benefits substantially from GRPO refinement.

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Case	Reference	v3 output (failure)	v5 output (recovered)
Base fresh idx 14	Điều tra vụ anh rể dùng dao đâm chết em vợ vào rạng sáng	Output assistant assistant assistant assistant assistant assistant ... (×128 tokens) Failure signature • Pure repetition loop — no content • ROUGE-2 = 0.00 • LenErr = 814.29% • R_acc = 0.00	Output Anh C. bị anh rể đâm chết tại nhà. Recovered properties • Short, content-bearing headline • Named entity retained • Budget restored • ROUGE-2 = 0.19
Base SFT-init idx 7	5 năm chưa xử xong vụ án "lạ" có hơn 30 luật sư bào chữa miễn phí	Output Bị cáo Tuyết bị cấp sơ thẩm 2 lần tuyên phạt 12 năm tù . Bị cáo Tuyết kháng cáo , nên 2 lần vụ án được đưa ra xét xử phúc thẩm và đều có chung 1 kết quả , huỷ án Failure signature • 245 words — punctuation-heavy blow-up • Meaningful start → punctuation flood • LenErr = 1189.47% • ROUGE-2 = 0.034	Output Vụ án "lạ" : Huỷ án 2 lần , bị cáo vẫn bị tạm giam 5 năm ! Recovered properties • Compact, accurate rewrite • Budget restored • Content preserved • ROUGE-2 = 0.22

Figure 4: Paired v3–v5 examples of the two most severe reward-hacking modes from VietNews. Each card shows a reference summary (gray), the failed v3 output (red), and the recovered v5 output (green), together with failure or recovery indicators. Left: repetition loop (Base fresh; assistant repeated 128 times). Right: punctuation-driven length explosion (Base SFT-init; 245 words of periods).

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A Appendix

A.1 Full Within-project Variant Tables

Tables 8–10 provide the full two-dataset view for all canonical Qwen3 project variants. These tables provide additional detail for the single-document evaluation suite.

Table 8: ROUGE-2 (%) across the two primary single-document datasets for the canonical Qwen3 project variants

Model	VietNews	WikiLingua
Base pretrained	12.13	3.18
Base + SFT	26.61	24.43
Base + fresh GRPO v5	20.73	9.71
Base + SFT + GRPO v5	<u>28.97</u>	26.35
Instruct pretrained	16.48	7.44
Instruct + SFT	26.70	23.93
Instruct + fresh GRPO v5	18.67	8.09
Instruct + SFT + GRPO v5	29.62	<u>25.59</u>

Table 9: Absolute summary length distance across the two primary single-document datasets for the canonical Qwen3 project variants

Model	VietNews	WikiLingua
Base pretrained	33.03	61.59
Base + SFT	5.73	14.18
Base + fresh GRPO v5	3.80	22.03
Base + SFT + GRPO v5	<u>1.16</u>	8.35
Instruct pretrained	4.01	23.42
Instruct + SFT	1.09	<u>9.87</u>
Instruct + fresh GRPO v5	2.95	16.12
Instruct + SFT + GRPO v5	1.52	10.25

Table 10: Relative length error (%) across the two primary single-document datasets for the canonical Qwen3 project variants

Model	VietNews	WikiLingua
Base pretrained	205.89	174.67
Base + SFT	31.19	24.72
Base + fresh GRPO v5	22.36	50.94
Base + SFT + GRPO v5	<u>6.43</u>	14.69
Instruct pretrained	24.50	57.47
Instruct + SFT	6.26	18.34
Instruct + fresh GRPO v5	17.08	38.67
Instruct + SFT + GRPO v5	9.24	<u>18.11</u>

A.2 Historical Comparison of Two GRPO Configuration Bundles

This appendix records the two GRPO configuration bundles available in the stored artefacts: v4 ($K = 4$, learning rate 5×10^{-7} , $\beta = 0.15$) and v5 ($K = 8$, learning rate 2×10^{-6} , $\beta = 0.04$). Because only two operating points were evaluated, three hyperparameters change jointly, and the sampling-based evaluation has no saved seed, this comparison is not a controlled hyperparameter ablation or a systematic search.

Table 11: Historical comparison of two explored GRPO configuration bundles on VietNews and WikiLingua macro averages. These observations do not identify the effect of any individual hyperparameter or establish a globally optimal configuration

Branch	v4 ROUGE-2	v5 ROUGE-2	v4 LenErr%	v5 LenErr%	v4 LenDist	v5 LenDist
Base fresh	10.93	15.22	91.67	36.65	26.70	12.90
Base SFT-init	26.25	27.66	<u>17.88</u>	10.56	<u>7.05</u>	4.80
Instruct fresh	12.39	13.38	33.98	27.88	11.65	9.50
Instruct SFT-init	<u>26.23</u>	<u>27.60</u>	13.09	<u>13.68</u>	6.00	<u>5.85</u>

Across the stored runs, v5 has higher ROUGE-2 in all four branches and lower length distance in all four branches. Relative length error is lower under v5 in three branches and slightly higher for Instruct SFT-init. These values are retained for transparency and hypothesis generation only, not as evidence for a causal configuration claim.

A.3 Sentence-control Results

Sentence-control results are retained here because the main paper focuses on ROUGE-2 and absolute summary length distance.

Table 12: Sentence-control metrics recomputed from stored generations. “Tol.” counts a prediction as correct when the sentence count is within ± 1 of the requested budget

Model	VN Exact%	VN Tol.%	VN MAE	WL Exact%	WL Tol.%	WL MAE
Base pretrained	73.95	89.55	0.550	15.4	32.4	2.910
Base + SFT	91.40	96.55	4.299	58.6	83.8	1.674
Base + fresh GRPO v5	91.80	<u>99.30</u>	0.089	34.2	72.8	1.226
Base + SFT + GRPO v5	93.85	<u>98.75</u>	0.316	65.6	95.6	0.470
Instruct pretrained	93.95	<u>99.40</u>	0.067	34.2	64.6	1.582
Instruct + SFT	<u>95.45</u>	98.90	<u>0.058</u>	88.0	97.0	0.204
Instruct + fresh GRPO v5	94.85	99.60	0.056	46.8	72.4	1.178
Instruct + SFT + GRPO v5	96.95	99.05	0.041	78.8	95.6	<u>0.340</u>

A.4 Dataset Processing Details

The final JSONL artefacts are produced by the project’s data processing and augmentation modules. The most important preprocessing rules are listed below.

- Shared preprocessing truncates sources to 8,000 characters and summaries to 1,500 characters, then replaces underscore-based segmentation markers with spaces.
- VietNews uses the first line of each file as the target headline, the sapo plus body as source text, and drops the last line as caption-like trailing content.
- WikiLingua flattens the source and target sentence lists into single strings.

A.5 Pipeline Configurations and Prompt Templates

This subsection summarizes the end-to-end pipeline configuration used by the paper. The source of truth is the project’s canonical configuration files, training scripts, evaluation pipeline, and audit documentation. Prompt examples are shown in Vietnamese Unicode and require XeLaTeX in this project setup, but they follow the same instruction format used throughout the pipeline.

Table 13: Prompt-construction rules used by the pipeline

Process	Prompt shape	Template policy	Stored outputs
SFT	system + user + assistant	Always uses the approximate control template family	messages plus metadata
GRPO train/val	system + user	Rotates three length templates and three sentence templates across prompts	prompt, reference, and metadata
Evaluation/test	system + user	Deterministic test prompt; template index fixed to 0	prompt, reference, and metadata

Table 14: Control-template families used in the pipeline

Signal	Template families	Notes
Length	<i>khoảng X từ; trong khoảng lo-hi từ; không quá X từ</i>	SFT uses only the first form; GRPO rotates all three; eval fixes the first form
Sentence	<i>khoảng X câu; trong khoảng lo-hi câu; không quá X câu</i>	Same rotation policy as the length templates

All prompt-bearing samples preserve metadata for the reward functions and analysis logs. The core metadata fields are target length, target sentence count, length requirement text, and sentence requirement text; the pipeline also tracks statistics such as source length.

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Canonical SFT configuration.

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Table 15: Canonical SFT configuration used by the paper

Setting	Canonical paper configuration
Backbone family	Qwen3-4B-Base or Qwen3-4B
LoRA adapter	$r = 32$, $\alpha = 32$, dropout 0.05; attention projections (query, key, value, output) and MLP projections (gate, up, down)
Sequence length	3072 on H200/A100; 2048 on V100
Training	$lr = 5 \times 10^{-5}$, cosine scheduler, $warmup_ratio = 0.1$, $epochs = 2.0$
Thinking suppression	Enabled
Batch calibration	Enabled before training
Checkpoint behavior	Model-family-specific output directory; resume from latest checkpoint when present

Table 16: SFT per-GPU starting presets used in the canonical training runs. Auto-calibration can adjust these values at runtime

GPU tier	Start batch	Grad. accum.	Effective batch
H200	10	2	20
A100	6	3	18
V100	2	4	8

The appendix tables intentionally follow the launch-script presets rather than the raw library defaults, because the launch scripts are the source of truth for the canonical paper runs.

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Canonical GRPO v5 configuration.

Table 17: Canonical GRPO v5 configuration used in the main paper results

Setting	Canonical paper configuration
Initialization	SFT-initialized warm start
Core v5 hyperparameters	$K = 8$, $lr = 2 \times 10^{-6}$, $\beta = 0.04$, $total_steps = 800$
Rollout decoding	Temperature 0.7, top- p 0.9, random sampling, max 80 new tokens, max sequence length 3072
Reward weights	Accuracy 0.5, length 0.3, sentence 0.2
Stability settings	Gradient checkpointing enabled, thinking mode suppressed, auto-calibration enabled
Repetition controls	Repetition penalty 1.0, n-gram blocking disabled

Table 18: GRPO per-GPU starting presets used in the canonical training runs. Auto-calibration can refine these values at runtime

GPU tier	Start batch	Grad. accum.	Effective prompts/update
H200	4	4	16
A100	2	8	16
V100	1	16	16

For reference only, the main ablation bundle changes are $v4$: $K = 4$, $lr = 5 \times 10^{-7}$, $\beta = 0.15$; $v3$ and $v4$ remain ablation settings and should not be read as the canonical appendix configuration.

Evaluation configuration.

Table 19: Frozen evaluation configuration used to generate the paper tables

Setting	Frozen evaluation protocol
Decoding	Temperature 0.3, top- p 0.9, random sampling, max 256 new tokens
Batching	Batch size 8
Prompt formatting	Chat template applied with thinking mode disabled when supported
Prompt truncation	Tokenizer truncates prompts to 3072 minus the maximum allowed new tokens
Evaluation inputs	Per-dataset test artefacts plus the combined test split
Metrics in main paper	ROUGE-2 and Length Distance; sentence-control kept in appendix
Randomness	One-shot low-temperature sampling; no evaluation seed was saved

SFT prompt example.

System: Bạn là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: khoảng 22 từ
- Số câu: khoảng 1 câu

Văn bản: <source article>

Assistant: <gold summary>

GRPO rollout prompt example.

System: Bạn là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: trong khoảng 18-26 từ
- Số câu: trong khoảng 1-2 câu

Văn bản: <source article>
reference = <gold summary>
meta.target_length = 22
meta.target_sentences = 1
meta.length_requirement = “trong khoảng 18-26 từ”
meta.sentence_requirement = “trong khoảng 1-2 câu”

Evaluation/test prompt example.

System: Bạn là một trợ lý AI chuyên tóm tắt văn bản tiếng Việt. Hãy tạo ra bản tóm tắt ngắn gọn, chính xác, tuân thủ đúng yêu cầu về độ dài và số câu.

User:

- Độ dài: khoảng 22 từ
- Số câu: khoảng 1 câu

Văn bản: <source article>
reference = <gold summary>
meta.dataset = “vietnews”

GRPO training and evaluation share the same prompt shape: both keep only the system and user turns, store the gold summary separately as *reference*, and preserve the control metadata in *meta*. The differences are that GRPO rotates all template families during rollout training, whereas evaluation fixes the approximate template, and SFT is the only stage that includes an assistant turn inside the serialized chat sample.

A.6 No-sentence-reward Ablation

The no-sentence ablation is informative but not decisive. Both sides of the comparison still use sampling without a saved evaluation seed, so the correct conclusion is *inconclusive* rather than a strong causal claim about sentence reward.

Table 20: No-sentence-reward ablation computed from a frozen evaluation snapshot. The differences are too small under the current sampling-based protocol to support a strong causal statement

Branch	w_{sent}	ROUGE-2	LenErr%	Sent. Exact%	Sent. MAE
Base fresh v4	0.2	13.88	76.44	59.42	1.136
Base fresh ablation	0.0	13.99	76.50	60.32	<u>1.163</u>
Base SFT-init v4	0.2	<u>26.71</u>	<u>20.82</u>	<u>73.39</u>	<u>2.009</u>
Base SFT-init ablation	0.0	26.85	17.21	73.65	1.452